2023 27th International Conference Information Visualisation (IV) | 979-8-3503-4161-4/23/$31.00 ©2023 IEEE | DOI: 10.1109/IV60283.2023.00048

Artificial Intelligence in Visual Analytics

Kawa Nazemi
Human-Computer Interaction and Visual Analytics
Darmstadt University of Applied Sciences, Darmstadt, Germany
kawa.nazemi@h-da.de

***Abstract*—Visual Analytics that combines automated methods with information visualization has emerged as a powerful approach to analytical reasoning. The integration of artificial intelligence techniques into Visual Analytics has enhanced its capabilities but also presents challenges related to interpretability, explainability, and decision-making processes. Visual Analytics may use artificial intelligence methods to provide enhanced and more powerful analytical reasoning capabilities. Furthermore, Visual Analytics can be used to interpret black-box artificial intelligence models and provide a visual explanation of those models. In this paper, we provide an overview of the state-of-the-art of artificial intelligence techniques used in Visual Analytics, focusing on both explainable artificial intelligence in Visual Analytics and the human knowledge generation process through Visual Analytics. We review explainable artificial intelligence approaches in Visual Analytics and propose a revised Visual Analytics model for Explainable artificial intelligence based on an existing model. We then conduct a screening review of artificial intelligence methods in Visual Analytics from two time periods to highlight recently used artificial intelligence approaches in Visual Analytics. Based on this review, we propose a revised task model for tasks in Visual Analytics. Our contributions include a state-of-the-art review of explainable artificial intelligence in Visual Analytics, a revised model for creating explainable artificial intelligence through Visual Analytics, a screening review of recent artificial intelligence methods in Visual Analytics, and a revised task model for generic tasks in Visual Analytics.**

***Index Terms*—Visual Analytics, Artificial Intelligence, Neural Networks, Clustering, Classification, Explainable AI, XAI, Similarity, Deep Learning, Survey**

I. Introduction

Artificial intelligence (AI) has emerged as a transformative technology with wide-ranging applications in various domains [1]–[4]. Visual Analytics has emerged as the science of analytical reasoning facilitated by interactive Visualization combining automated methods with information visualization [5, 6].

AI techniques have been integrated into Visual Analytics to enhance its capabilities and provide advanced analytical capabilities [7]. However, integrating AI in Visual Analytics also presents challenges, such as the need for interpretable and explainable AI models, the management of biases, and best-fitting methods to either explain the underlying models or enable an appropriate decision-making process [8]. Explainability and interpretability are critical in Visual Analytics, as analysts need to understand how AI models arrive at their results to trust and make informed decisions based on them [2]. The terms “explainability” and “interpretability” are often used interchangeably in the field of AI, but they can have distinct meanings [9]–[11]. While there is no universally agreed-upon definition, we try to provide an overview of the general distinctions. We use the term ”explainability” for the ability of an AI system to provide understandable explanations or justifications for its decisions or predictions, focusing on providing a human-readable explanation of the reasoning or logic behind a model’s output. This can involve presenting the key factors, features, or rules influencing the system’s decision-making process. Explainability aims to bridge the gap between the internal workings of AI models and human comprehension, allowing users to understand the decision-making process [10]–[12]. We use the term “interpretability” for the degree to which the internal mechanisms or operations of an AI model can be understood and analyzed by humans. It emphasizes the transparency and comprehensibility of the system’s internal workings. It aims to provide insights into how an AI model processes inputs, transforms them, and produces outputs. This understanding can help assess the system’s reliability, robustness, and trustworthiness. Interpretability focuses on making AI models more transparent and enabling human experts to analyze and validate the model’s behavior [2, 9]. Explainability and interpretability are not mutually exclusive but rather interconnected concepts. A system’s interpretability can contribute to its explainability, as understanding the inner workings of the system enables meaningful explanations. Conversely, explanations provided by an AI system can enhance interpretability by making the system’s outputs more understandable and facilitating further analysis.

Explainable Artificial Intelligence (XAI) emerged from the need to understand the underlying models. There exist a large number of XAI methods that lead to understanding even so-called black-box models and interpreting them through Visual Analytics. However, no simple and appropriate model exists to integrate AI in Visual Analytics in an abstract way leading to understanding the underlying models and methods, although Visual Analytics integrates more and more AI methods to enable complex problem-solving processes. Complex problem-solving and analytical reasoning are the main goals of Visual Analytics [1, 5]–[7, 13]. Therefore it is necessary to investigate the role of AI in Visual Analytics, particularly in solving analytical tasks. As Visual Analytics may integrate any kind of automated analysis method, the problem space is not limited to AI. However, AI in Visual Analytics leads to further challenges besides a transparent and interpretable model. One main challenge is to rework the task models. As Visual Analytics in combination with AI leads to solving more analytical tasks,

2375-0138/23/$31.00 ©2023 IEEE
DOI 10.1109/IV60283.2023.00048

it is necessary to provide a generic task model for Visual Analytics, including those aspects and challenges that arise through the combined use of AI and Visual Analytics since Visual Analytics is not only to predict or interpret as stated often in recent research [1]. Generic task-solving models [14] should be available to promote the value of Visual Analytics.

We provide in this paper an overview of the state-of-the-art of AI techniques used in Visual Analytics. Therefore, we investigate two different aspects of AI in Visual Analytics: (1) Explainable AI in Visual Analytics and (2) Human knowledge generation processes in Visual Analytics that leads to solving analytical tasks. In the first case, we investigate the use of Visual Analytics to explain or interpret AI, while the scope of the second case is how humans can generate knowledge, e.g., to make decisions through visualizations, where AI processes the data. We start with a state-of-art review of XAI in Visual Analytics that should give an idea about current approaches and methods that lead to understanding particularly black-box AI models. Based on the review, we provide a Visual Analytics model for XAI using the proposed Visual Analytics Model of Keim et al. [6]. Thereafter, we provide a screening review in several databases to illustrate the number of Visual Analytics approaches using AI methods. The main goal here is to outline those techniques that emerged recently. We, therefore, divide our review into two time periods: (1) The period from 2005 to 2018 as the most influential work on Visual Analytics [5] appeared in 2005 and (2) from 2019 to April 2023 (time of writing this paper). The screening review will allow a comparison between the time periods and enable the definition of generic tasks. We will use the task model of Munzner [14] to provide a revised task model consisting of actions and targets. Our main contributions are (1) a state-of-the-art review of XAI in Visual Analytics, (2) a revised model to enable creating XAI through an abstract and simplified Visual Analytics model, (3) a screening review of AI methods and approaches in Visual Analytics subdivided into two time periods, and (4) a revised task model for generic tasks in Visual Analytics consisting of actions and targets.

II. Visual Analytics for Explainable Artificial Intelligence

Explainable artificial intelligence aims to provide understandable AI results for end-users. XAI techniques focus on developing machine learning methods that offer explainable, trustworthy, and comprehensible rationales for decisions made by black-box models. Black-box models are characterized as systems with an opaque and uninterpretable internal functioning or logic [15]. Machine Learning (ML) algorithms enhance their performance by improving a function to minimize a loss function, which measures how well it predicts outputs for unseen inputs [1, 16]. Interpretations of algorithms such as regressions, decision trees, k-Nearest Neighbor, or Bayesian classifiers can be derived from their model parameters, structure, and rules [1, 17, 18]. However, these methods require better accuracy despite providing transparency and explanations of their predictions [17]. To address this, researchers have utilized more powerful models, such as neural networks (NN), support vector machines (SVM), and ensemble methods, which have performed well in various domains [19]. However, these models are commonly considered as black-box models because they are difficult to interpret and explain. In particular, NN-based models pose a challenge to interpretability due to their complex structure, which includes hyperparameters, hidden layers, and neurons, that can obstruct transparency [1, 17, 20]. This increased complexity in NN-based models can make it difficult for domain users to understand how they arrive at their predictions or results. Black-box models do not provide users with a clear understanding of the inner behavior of those models and lead to uncertainty of the output results.

A deep neural network model is composed of interconnected node layers, each with adjustable weights [16]. Feature values are forwarded from input layers to hidden layers with random initial weights. Hidden layers facilitate connections between input and output by performing nonlinear transformations using activation functions [21]. The values of hidden nodes are obtained by multiplying the weights of previous layer nodes and adding them up [22]. The output layer is the sum of the last hidden layer nodes obtained through an activation function [1, 16, 21]. Optimization algorithms update the weights and compare the obtained output to actual values, minimizing loss through backpropagation [21]. Deep neural networks excel in modeling complex nonlinear systems through their hidden layers and non-linear activation functions. Increasing the number of hidden layers allows the modeling of more complex relationships in high-dimensional data [16, 21]. Additionally, deep neural networks can automatically extract features from raw data, eliminating the need for feature engineering tasks [1]. Despite their superior modeling ability, the complex structure of NN models renders them difficult to interpret compared to other models. To address this challenge, many explainable artificial intelligence methods have been developed [1, 10, 23]–[26] to explain the according black-box models. By achieving these goals, model designers can more easily identify potential issues with the computational process while also receiving insights for improving models.

Besides understanding the black-box models, a main research challenge is reducing the effort needed to build and fine-tune those models [25]. Explaining AI models leads to a better understanding of the black-box models, whereas improving models by fine-tuning through visual interfaces reduces the effort and makes the entire process more comprehensible [27]. Interactive visualization may show the process of a neural network over time [28] to enable understanding of the process, simplify the tuning process of classifier weights [27, 29] or error-driven feature ideation for classification problems [30]. Brooks et al. proposed with *FeatureInsight* such an error-driven feature ideation system based on examining classifier errors and comparing errors [30]. FeatureInsight is a workflow-based system for feature ideation that employs two recommended approaches: analyzing classifier errors and conducting comparisons. It allows users to inspect misclassified documents and prioritize errors based on the accuracy of document predictions. By providing side-by-side comparisons of misclassified documents and their contrasting counterparts, FeatureInsight aids users in identifying discriminative characteristics and generating new features to improve classifier performance [30].

Explainable AI can be defined and categorized in several ways. One common differentiation is local versus global XAI [1, 10, 31, 32]. Local XAI refers to the process of explaining the decision-making for a specific prediction or decision. It aims to provide insight into how the model arrived at its decision by identifying which features or inputs were most important in determining the output. Because the decision boundary of a black-box model may be highly complex across the entire data space, but in the neighborhood of a given data point, it is more likely to be clear and easily interpretable. Therefore, an interpretable model can be utilized to capture this boundary effectively [31, 32]. Global XAI refers to the process of explaining the decision-making of a model at a high-level, such as over an entire dataset or population. It aims to provide an overall understanding of how the model works and why it makes certain decisions [1]. XAI can be further differentiated through model-specific vs. model-agnostic [15], intrinsic (rule-based) [33] vs. post-hoc explanations [23], qualitative vs. quantitative [10], and human-in-the-loop [34] vs. automated. Model-specific XAI methods are designed for a particular type of machine learning model, while model-agnostic methods can be applied to any model [1, 15]. Rule-based or intrinsic methods involve building interpretable models directly by the model itself [33], while post-hoc XAI methods provide explanations for existing black-box models [10, 23, 26]. Qualitative methods provide explanations in natural language or visual form [2, 10, 32, 35], while quantitative methods provide numerical or statistical measures of the model's behavior [3, 9, 10, 36]. Human-in-the-loop XAI involves incorporating human feedback into the model-building process [37]–[39], while automated XAI methods are designed to work independently of human input [2, 40].

We found that the most used XAI methods are local, model-agnostic, post-hoc, quantitative, and automated. The most used approaches in Visual Analytics are LIME [10] (open source [1]) and SHAP [36] (open source [2]). Besides LIME and SHAP, two popular approaches are rarely used in Visual Analytics to explain models: ANCHOR [12] (open source [3]) and LRP [26] (open source [4]). LIME learns an interpretable model locally around the selected instance by using a surrogate model such as linear or tree-based models. LIME generates samples around the selected instance as a black star by perturbing that changes the selected instance to create similar instances, illustrated as gray stars [10]. SHAP treats features as team members of a game. It calculates the relative contribution of each feature to the individual prediction, corresponding to each team member's contribution to winning the game [36].

[1]LIME: https://github.com/marcotcr/lime
[2]SHAP: https://github.com/slundberg/shap
[3]ANCHOR: https://github.com/marcotcr
[4]LRP: https://github.com/sebastian-lapuschkin/lrp_toolbox

Visual Analytics is, per definition, a "qualitative" way for XAI. One can assume that Visual Analytics approaches will be most helpful through human-in-the-loop methods of XAI. Let us see one simple example of (Munro) Monarch [38] for annotating data in a neural network. Monarch proposes an iterative process for a simple human-in-the-loop machine learning system. Initially, a random sample of unlabeled items are annotated to set aside as evaluation data. Followed by labeling the first items to be used for training data, also starting with a random selection. Hereafter active learning is used to sample low-confidence or outliers items [38, p. 32]. Figure 1 illustrates the process in a simplified way, whereas the steps are not illustrated.

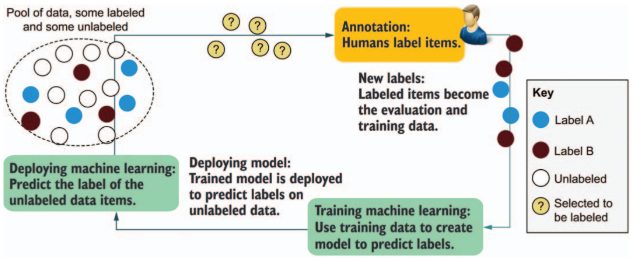

Fig. 1. A simple architecture of a human-in-the-loop machine learning system [38, p. 25]. Starting with a pool of data, humans are labeling the data items. These become training and evaluation (test) data that lead to training the model and predicting labels. The trained model is deployed to predict unlabeled data items.

One key advantage of Visual Analytics is its ability to enable human-in-the-loop decision-making. While AI algorithms can be highly effective at processing and analyzing data, they may lack the ability to incorporate human expertise and intuition into the decision-making process. With Visual Analytics, human experts can interact with and guide the AI algorithms, providing feedback and insights that can improve the accuracy and relevance of the analysis. The human-in-the-loop process is not necessarily composed of qualitative methods. Moreover, model-agnostic quantitative XAI methods that allow parameter refinement should and can be included in the Visual Analytics process to enable experts an in-depth analysis and real-time model-interaction approach with the underlying models. One of the first Visual Analytics models was proposed by Keim et al. [6] as a rhombus combining data to visualization (mapping), data to models (data mining), and models and visualizations to knowledge with a feedback loop from knowledge to data [6, p. 10]. Thereby a direct coupling of visualization and models was proposed with parameter refinement in models and user interaction in visualizations. While the model of Keim et al. was surely a main step in the direction of combining artificial intelligence with interactive visualizations and allowing parameter refinements, the technological advancements lead to revisiting the model and refining it to integrate the existing approaches of XAI in the model for a better understandability and easier refinement of model parameters. We, therefore, revised the model of Keim et al. [6] as illustrated in Figure 2.

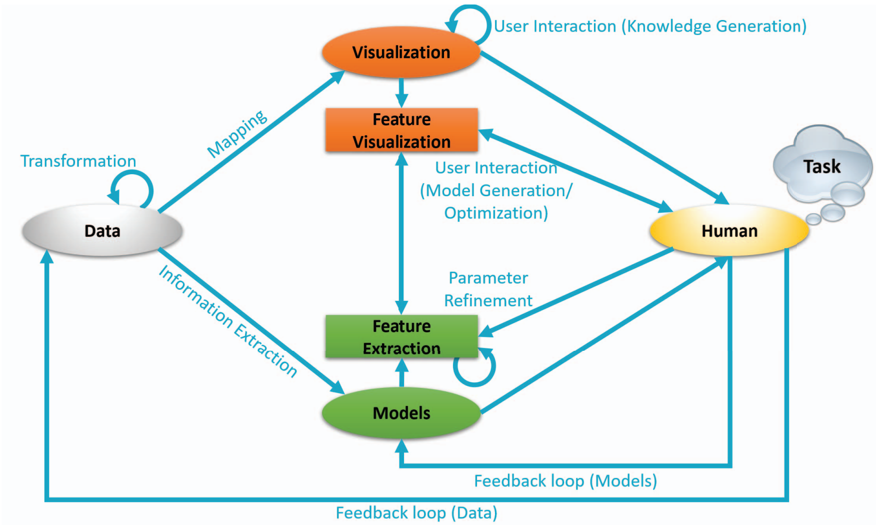

Fig. 2. A revised Visual Analytics model with a human-in-the-loop method in Artificial Intelligence, based on the original model of Keim et al. [6, p. 10]

Our model starts with data that are transformed and mapped to visual structures [41], while data features are extracted through any kind of information extraction method for the ML/AI models. A model-agnostic, local, quantitative, post hoc method extracts the feature of the models quantitatively and provides a second level of interactive visualizations. Humans (instead of knowledge) are enabled to interact with the main visualization to gain knowledge or perform analytical tasks. In parallel, the features of the models are visualized to enable both an insight into the model that generated the information for humans and model optimization through interactive visual representations. The feedback loop is enhanced to enable a real parameter refinement (feedback loop - models) besides the feedback loop to data. The main interactive visualization still allows knowledge generation and analysis and enable, through a variety of interaction techniques an in-depth analysis.

III. Artificial Intelligence for Extracting Knowledge through Visual Analytics

Visual Analytics enables the exploration and understanding of complex data sets through interactive visualizations. The ability to analyze and interpret large amounts of data is a critical component of AI, and visual analytics offers a way to make this process more efficient and effective. Visual Analytics in AI enables gaining insights into data patterns and trends [13] that might be difficult to discern through traditional analytical methods, allowing users to quickly identify patterns, anomalies, and relationships within the data. Visual Analytics provides per definition [5]–[8] to interactively explore and manipulate data in real-time, enabling users to experiment with different data visualization techniques, filtering and aggregation methods, and other analytical tools [42]. This allows users to gain a deeper understanding of the data and to identify insights that might not have been apparent through traditional analytical methods. This is particularly because of two main reasons: Visual Analytics provides juxtaposed, superimposed interactive visualizations or a combination of both, and Visual Analytics enables solving various tasks [14]. Munzner proposed a task abstraction model to summarize and categorize possible tasks that can be solved through Visual Analytics [14]. She stated that a task could be defined as a pair of actions and targets. Thereby actions are categorized into “analyze”, “search”, and “query”, and targets are “all data”, “attributes”, “network data”, and “spatial data”. According to Munzner an analytical task could be “discover trends“. Thereby, “discover” is a task within ”analyze” and “trends” is a target within the category “all data”. Figure 3 illustrates the task model proposed by Munzner [14].

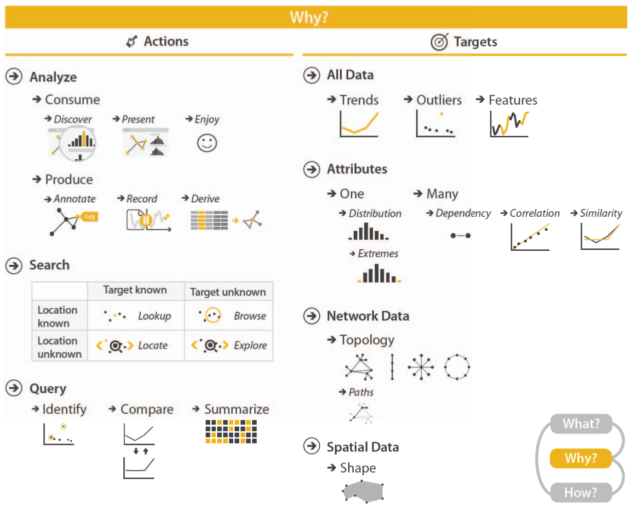

Fig. 3. The task model for visualizations and Visual Analytics proposed by Munzner [14, p. 42]

TABLE I

RESULTS OF OUR SCREENING REVIEW IN DIFFERENT DATABASES: GREEN REFERS TO A SIGNIFICANTLY INCREASING AMOUNT OF PUBLICATIONS, WHILE RED REFERS TO A DECREASING AMOUNT.

	Web of Science 2005-2018	**Web of Science** 2019-2023	**IEEE XPlore** 2005-2018	**IEEE XPlore** 2019-2023	**ACM DL** 2005-2018	**ACM DL** 2019-2023	**Springer** 2005-2018	**Springer** 2019-2023	**Total**
Clustering / Similarity	**230**	**219**	**345**	**164**	**1,958**	**1,968**	**1,747**	**1,958**	**8,589**
Classification	70	99	113	106	970	472	1,148	1,631	4,879
Neural	21	88	31	160	133	300	436	1,274	2,443
Natural Language Processing	12	26	26	47	125	133	417	838	1,624
Topic Modeling	11	19	12	20	61	45	147	184	499
Segmentation	22	29	33	29	415	274	320	805	1,927
Data Mining	78	64	343	141	445	203	1,414	1,414	4,102
Total	444	544	903	667	4,107	3,395	5,629	8,104	24,063

For solving tasks in Visual Analytics, using methods and models of artificial intelligence is crucial [7]. There exist many methods and algorithms that enable the knowledge generation process of humans and allow them to interpret huge amounts of data or predict possible future scenarios [43]. While neural networks are used as underlying models in recent Visual Analytics approaches increasingly often [4, 44, 45], those approaches mostly investigate the neural networks as the subject of analysis. Xuan et al. proposed a system to compare CNNs through a Visual Analytics system [4]. Their system allows inspecting a single CNN in-depth and comparing more than two CNN models [4]. Therefore they set up a three-step workflow of (1) information overview for general (quantitative) information of the models, (2) task customization to support visual explanations and comparison rules, and (3) presenting coordinated visualization and qualitative information for multi-model comparison [4]. The main task that can be solved here is the analytical "comparison" tasks through a similarity matrix [4]. Wang et al. proposed with *DR*LIVE a Visual Analytics system to explore, interpret and diagnose Recurrent Neural Network (RNN) based Deep Reinforcements Learnings (DRLs) in games [45]. Their system enables game episode exploration, RNN hidden and cell state examination, and mode perturbation through visual interaction [45]. DRL agents are first explored visually, followed by discovering interpretable RNN cell of those DLRs by prioritizing RNN hidden/cell states with a set of metrics. Finally, the DRL models are diagnosed by interactively perturbing their inputs [45]. These two examples should outline that Visual Analytics is considering neural networks and advanced learning approaches, e.g., DRL, as a subject of investigation and not only as a model that is used to analyze data. Although neural networks and advanced machine learning methods are applied more frequently in Visual Analytics, the "common" models and methods are still used mostly for analyzing data and gaining knowledge for the different tasks [14] in Visual Analytics.

To provide a revised and up-to-date task model for Visual Analytics, we conducted a screening review of the approaches and methods used in Visual Analytics. Our screening review compares methods and models integrated into Visual Analytics systems and approaches between 2005 and 2018 with those published between 2019 and 2023. We chose 2005 as a starting point of our research due to the fact that the term "Visual Analytics" was established through Thomas and Cook in 2005 [5]. The research included the following databases and publishers: *Web of Science*, *IEEE Xplore*, *ACM Digital Library (ACM DL)*, and *Springer (Springer Link)*.

We used Boolean operators and specific publishers' advanced search functionality. The search queries were performed using either a given time period search in the publishers' databases or by facetting the time period after the search. The term Visual Analytics was always set in quotes, and the second term was combined with a logical `AND`, e.g., `"Visual Analytics" AND Classification`. In cases of two similar technologies, we used the logical `OR` operator, e.g., `"Visual Analytics" AND (Clustering OR Similarity)`. The terms were extracted through our Visual Analytics system for emerging technologies [8, 13] and investigation of surveys in the field [1, 7, 15, 16, 20, 35, 43]. For any kind of neural network and deep learning, we used the term "neural" since this term occurs in the related literature. Since clustering and similarity techniques are very similar and the terms are commonly used for both, we combined these terms as one approach.

Table I illustrates the results of our screening review. We highlighted in our review significant changes according to the number of publications through a light red color for a decreasing amount of published papers and green for an increased number. The comparison is performed between two time periods, which are significantly different: 14 years (2005-2018) and just four and a half years (2019-2023, April). The goal was to highlight the emerging recent methods in Visual Analytics.

Overall the most used method in Visual Analytics since 2005 is "Clustering / Similarity" with 8,589 publications and 4,309 publications between 2019 and April 2023. Significant changes are only visible in the IEEE Xplore database with a decreasing number of publications in recent years. *Clustering* algorithms are used to group similar data points [46]. The results of clustering algorithms like k-means and k-nearest neighbors are often visualized using scatter plots with dots colored according to their cluster [47]. To avoid dimensionality reduction, some approaches [47] use parallel sets, where ribbons connect axes representing features to show each cluster's proportional distribution of feature combinations. Similarity measures, e.g., Okapi, Cosine, Jaccard, BM25, or Kullback-Leibler, are commonly used in Visual analytics to group data points too [48]. Besides classical statistical similarity measures, Self Organizing Maps (SOM) [49] are often used [46, 50]. SOM is a type of neural network learning algorithm that iteratively trains a network of prototype vectors to represent a set of input data vectors [46, 49]. The network is typically organized as a 2-dimensional regular grid. During training, the algorithm iterates over the input data vectors, identifying the best-matching prototype vector and then adjusting both the best-matching prototype and its neighboring prototype vectors toward the input vector [46, 49]. SOMs organize trajectory data in an unsupervised mode [46, 50] through grid visualizations, e.g., density maps, distances orders, or u-matrix [50]. In recent works, similarity algorithms are coupled with natural language processing approaches to visualize, e.g., similar document collections [51].

The second most used method in Visual Analytics since 2005 is "Classification" with 4,879 publications and 2,308 between 2019 and April 2023. Significant changes are only visible in the ACM DL with a decreasing number of approaches. In the last four years 472 papers were stored in ACM DL. In visual analytics, classification algorithms such as random forest and k-nearest neighbors are commonly used to assign data points to different classes [30, 44]. The performance of these algorithms is often evaluated and visualized using a confusion matrix, which provides an overview of the accuracy and misclassification of the predicted classes [52]. Visualization techniques, such as using colors to represent different classes, are often employed to enhance the interpretability of the classification results [26, 33, 44, 52].

The third most used method in Visual Analytics since 2005 is "Data Mining" with 4,102 publications and 1,822 between 2019 and April 2023. Significant changes are only visible in the IEEE Xplore and ACM DL, where the number of publications is halved. Data mining is extracting meaningful patterns, trends, or insights from large and complex datasets [13, 53]. It involves using various techniques and algorithms to discover hidden patterns, relationships, and knowledge from data, to make informed decisions, predict future outcomes, or solve complex problems [8, 13]. Data mining typically involves several steps, including data collection, data preprocessing, exploratory data analysis, model building, model evaluation, and interpretation of results [13]. Common data mining techniques include regression, association rule mining, sequential pattern mining, anomaly detection and also, classification, and clustering. These techniques utilize statistical, machine learning, and pattern recognition algorithms to uncover patterns and relationships within the data [6, 54]. We chose data mining as a category of investigation since the first Visual Analytics models considered data mining as the main step for model building [6].

The only term that increased significantly in all databases between 2019 and April 2023 is the combined use of "neural" and Visual Analytics. The term was chosen to consider any type of neural network. We did not investigate "deep learning" separately since commonly the term neural appears in those publications too, and the main ambition was to find out if neural networks are recently used in Visual Analytics as models. We already described neural networks and deep learning in Section II. It is important to recapitulate that neural networks are black-boxes that are the primary method of investigation in XAI.

We further investigated the use of natural language processing (NLP) in Visual Analytics. In a first step, we searched for the term NLP and found that particular topic modeling (TM) approaches [55] often do not contain NLP as a term. Therefore, we performed a dedicated search for topic modeling complimentary. With overall 1,624 publications containing NLP and 499 containing TM, the use of NLP methods in Visual Analytics is slightly increased in all databases and significantly increased in the Springer database. In visual analytics, NLP techniques are used to analyze and extract information from text data [13]. NLP techniques are used in text mining, sentiment analysis, entity recognition, topic modeling, summarization, and trend detection from text [8, 13, 53]. Integrating NLP techniques in Visual Analytics enables gaining deeper insights from textual data through interactive visualizations and exploratory data analysis [13].

The number of publications in Segmentation in Visual Analytics decreased in ACM DL and increased in the Springer database. With 1,137 publications between 2019 and April 2023, segmentation still plays an important role in Visual Analytics approaches. Segmentation refers to the process of dividing an input data set, such as an image, video, or audio, into meaningful and distinct segments or regions based on certain criteria or properties [56]. It is a fundamental task in computer vision and image processing. Segmentation can be performed using different approaches, including traditional methods based on handcrafted features and modern deep learning techniques that mostly leverage convolutional neural networks (CNNs) [56].

Our screening review reveals that AI methods, particularly neural networks, are used more frequently in Visual Analytics. This fact leads to a rethinking of the task model provided by Munzner [14]. In our revised task model, we applied the approach of "actions" and "targets" that lead to more generalized tasks, while the limitations are that specific tasks, e.g., "making decisions" can not be applied as such a generic task since making would not apply to "Extremes" or "Outliers". Figure 4 illustrates our revised task model based on the approach of Munzner [14]. We revised both, actions and targets and integrated XAI tasks too in the lower part of the model. Our main ambition was to create a generic task model that includes XAI and knowledge generation tasks in Visual Analytics. The task model is created through a rigorous screening of the recent approaches. Whereas some approaches describe the approach but not the tasks that are solved with

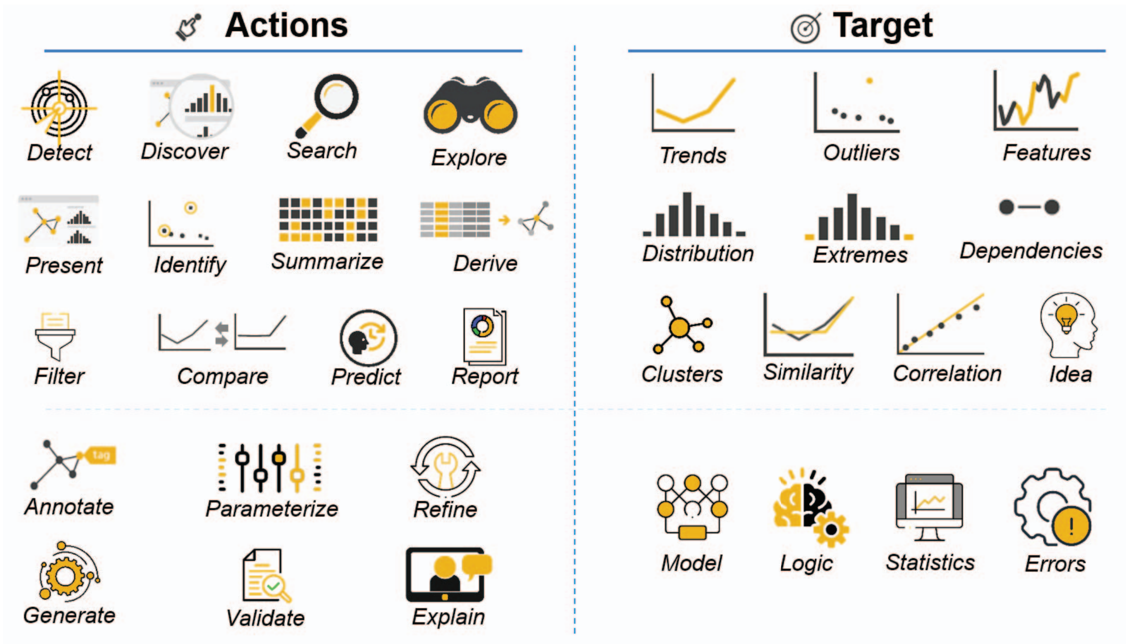

Fig. 4. Our revised task model for Visual Analytics based on the actions and target approach by Munzner [14, p. 42]. A task can be defined as a pair of actions and targets. Each of the actions on the left side can be combined with each of the targets on the right side. Although the lower part starting with "Ànnotate" on the left side and "Model" on the right side are more related to explaining, optimizing, and validating AI-models.

those approaches. We derived from the entire search results actions and targets that can be applied as a pair of actions and targets according to Munzner [14]. Although we did not investigate error detection of AI models in Visual Analytics, this topic is of growing interest in the community. Actions, such as explain, generate or validate may occur as exclusive XAI tasks, but those can be applied to any target, e.g., generate clusters or explain correlations.

IV. Conclusions

We provided in this paper an overview of the state-of-the-art review in AI using Visual Analytics. Therefore, we investigated both aspects of AI in Visual Analytics: (1) XAI in Visual Analytics and (2) Human knowledge generation processes in Visual Analytics that leads to solving various tasks. We started with a state-of-art review of XAI in Visual Analytics that gave an idea about current approaches and methods that lead to understanding particularly black-box AI models. Based on the review, we provided a Visual Analytics model for XAI using the proposed Visual Analytics model. Further, we provided a screening review in several databases to illustrate the number of AI-based Visual Analytics approaches. The main goal here was to outline those techniques that emerged recently. We, therefore, divided our review into two time periods: (1) The period from 2005 to 2018 as the most influential work on Visual Analytics [5] appeared in 2005 and (2) from 2019 to April 2023 (time of writing this paper). The screening review facilitated a comparison of different time periods and enabled the identification of common tasks. Despite not normalizing the two time periods, it was evident that neural networks, natural language processing, and segmentation were more frequently employed in recent Visual Analytics systems compared to the period between 2005 and 2018. However, to analyze the declining number of methods used in Visual Analytics, it will be necessary to normalize the total number of publications in the field, a task that will be addressed in future research. This finding further emphasizes that the combination of visualizations and NLP methods is both highly emerging and widely utilized. Furthermore, we presented an updated task model based on Munzner's work [14]. By adapting the task model to encompass current challenges and objectives in the field of Visual Analytics. We refined the model to include specific actions and targets associated with the tasks addressed in Visual Analytics.

Acknowledgment

This work was partially funded by the Hessian Center for AI (hessian.AI), Germany.

References

[1] G. Alicioglu and B. Sun, "A survey of visual analytics for explainable artificial intelligence methods," *Computers & Graphics*, vol. 102, pp. 502–520, feb 2022.
[2] C. Molnar, *Interpretable Machine Learning A Guide for Making Black Box Models Explainable*. Independently Published, 2022.
[3] A. Stevens, J. D. Smedt, and J. Peeperkorn, "Quantifying explainability in outcome-oriented predictive process monitoring," in *Lecture Notes in Business Information Processing*. Springer, 2022, pp. 194–206.
[4] X. Xuan, X. Zhang, O.-H. Kwon, and K.-L. Ma, "VAC-CNN: A visual analytics system for comparative studies of deep convolutional neural networks," *IEEE TVCG*, pp. 1–1, 2022.

[5] J. J. Thomas, *Illuminating the Path*. National Visualization and Analytics Ctr, 2005.
[6] D. Keim and F. Kohlhammer, Jörn; Ellis Geoffrey; Mansmann, Eds., *Mastering the information age : solving problems with visual analytics*. Goslar : Eurographics Association, 2010.
[7] B. Kovalerchuk, R. Andonie, N. Datia, K. Nazemi, and E. Banissi, "Visual knowledge discovery with artificial intelligence: Challenges and future directions," in *Studies in Computational Intelligence*. Springer International Publishing, 2022, pp. 1–27.
[8] K. Nazemi, D. Burkhardt, and A. Kock, "Visual analytics for technology and innovation management," *Multimedia Tools and Applications*, vol. 81, no. 11, pp. 14 803–14 830, may 2021.
[9] F. Doshi-Velez and B. Kim, "Towards a rigorous science of interpretable machine learning," 2017.
[10] M. T. Ribeiro, S. Singh, and C. Guestrin, ""why should i trust you?"," in *Proceedings of the 22nd ACM SIGKDD*. ACM, aug 2016.
[11] Z. C. Lipton, "The mythos of model interpretability," 2016.
[12] M. T. Ribeiro, S. Singh, and C. Guestrin, "Anchors: High-precision model-agnostic explanations," *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 32, no. 1, apr 2018.
[13] K. Nazemi and D. Burkhardt, "Visual analytics for analyzing technological trends from text," in *2019 23rd International Conference Information Visualisation (IV)*, Jul. 2019, pp. 191–200, best Paper Award.
[14] T. Munzner, *Visualization Analysis and Design*. A K Peters/CRC Press, 2014.
[15] A. Das and P. Rad, "Opportunities and challenges in explainable artificial intelligence (xai): A survey," arXiv, 2020.
[16] R. Garcia, A. C. Telea, B. C. da Silva, J. Tørresen, and J. L. D. Comba, "A task-and-technique centered survey on visual analytics for deep learning model engineering," *Computers & Graphics*, vol. 77, pp. 30–49, dec 2018.
[17] E. Dağlarli, "Explainable artificial intelligence (xAI) approaches and deep meta-learning models," in *Advances and Applications in Deep Learning*. IntechOpen, 2020.
[18] F. Emmert-Streib, O. Yli-Harja, and M. Dehmer, "Explainable artificial intelligence and machine learning: A reality rooted perspective," *WIREs Data Mining and Knowledge Discovery*, vol. 10, no. 6, jun 2020.
[19] A. Chatzimparmpas, R. M. Martins, I. Jusufi, K. Kucher, F. Rossi, and A. Kerren, "The state of the art in enhancing trust in machine learning models with the use of visualizations," *CGF*, vol. 39, no. 3, pp. 713–756, 2020.
[20] Y. Liang, S. Li, C. Yan, M. Li, and C. Jiang, "Explaining the black-box model: A survey of local interpretation methods for deep neural networks," *Neurocomputing*, vol. 419, pp. 168–182, jan 2021.
[21] B. D. Ripley, *Pattern Recognition and Neural Networks*. Cambridge University Press, jan 1996.
[22] C. B. Azodi, J. Tang, and S.-H. Shiu, "Opening the black box: Interpretable machine learning for geneticists," *Trends in Genetics*, vol. 36, no. 6, pp. 442–455, jun 2020.
[23] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," *International Journal of Computer Vision*, vol. 128, no. 2, pp. 336–359, oct 2019.
[24] N. Mehdiyev and P. Fettke, "Explainable artificial intelligence for process mining: A general overview and application of a novel local explanation approach for predictive process monitoring," in *Studies in Computational Intelligence*. Springer, 2021, pp. 1–28.
[25] N.-C. Chen, "Using visual analytics to support artificial intelligence development," Ph.D. dissertation, University of Washington, 2019.
[26] S. Bach, A. Binder, G. Montavon, F. Klauschen, K.-R. Müller, and W. Samek, "On pixel-wise explanations for non-linear classifier decisions by layer-wise relevance propagation," *PLOS ONE*, vol. 10, no. 7, p. e0130140, jul 2015.
[27] A. Kapoor, B. Lee, D. Tan, and E. Horvitz, "Performance and preferences: Interactive refinement of machine learning procedures," *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 26, no. 1, pp. 1578–1584, sep 2021.
[28] D. Smilkov, S. Carter, D. Sculley, F. B. Viégas, and M. Wattenberg, "Direct-manipulation visualization of deep networks," 2017.
[29] J. Talbot, B. Lee, A. Kapoor, and D. S. Tan, "EnsembleMatrix," in *Proceedings of the SIGCHI*. ACM, apr 2009.
[30] M. Brooks, S. Amershi, B. Lee, S. M. Drucker, A. Kapoor, and P. Simard, "FeatureInsight: Visual support for error-driven feature ideation in text classification," in *2015 IEEE VAST*. IEEE, oct 2015.
[31] M. Setzu, R. Guidotti, A. Monreale, F. Turini, D. Pedreschi, and F. Giannotti, "Glocalx – from local to global explanations of black box ai models," 2021.
[32] R. Guidotti, A. Monreale, F. Giannotti, D. Pedreschi, S. Ruggieri, and F. Turini, "Factual and counterfactual explanations for black box decision making," *IEEE Intelligent Systems*, vol. 34, no. 6, pp. 14–23, nov 2019.
[33] B. Letham, C. Rudin, T. H. McCormick, and D. Madigan, "Interpretable classifiers using rules and bayesian analysis: Building a better stroke prediction model," *The Annals of Applied Statistics*, vol. 9, no. 3, 2015.
[34] I. Lage, A. S. Ross, B. Kim, S. J. Gershman, and F. Doshi-Velez, "Human-in-the-loop interpretability prior," 2018.
[35] F. Xu, H. Uszkoreit, Y. Du, W. Fan, D. Zhao, and J. Zhu, "Explainable AI: A brief survey on history, research areas, approaches and challenges," in *Natural Language Processing and Chinese Computing*. Springer, 2019.
[36] S. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," 2017.
[37] P. J. Phillips, C. A. Hahn, P. C. Fontana, A. N. Yates, K. Greene, D. A. Broniatowski, and M. A. Przybocki, "Four principles of explainable artificial intelligence," Tech. Rep., sep 2021.
[38] R. Munro, *Human-In-the-Loop Machine Learning*. Manning Publications Company, 2020.
[39] D. Hadfield-Menell, A. Dragan, P. Abbeel, and S. Russell, "Cooperative inverse reinforcement learning," 2016.
[40] E. Real, C. Liang, D. R. So, and Q. V. Le, "Automl-zero: Evolving machine learning algorithms from scratch," 2020.
[41] S. K. Card, J. D. Mackinlay, and B. Shneiderman, *Readings in Information Visualization: Using Vision to Think.*, 1st ed. Morgan Kaufmann, 1999.
[42] N. Andrienko, G. Andrienko, G. Fuchs, A. Slingsby, and C. Turkay, *Visual Analytics for Data Scientists*. Springer, 2020.
[43] J. Ooge, G. Stiglic, and K. Verbert, "Explaining artificial intelligence with visual analytics in healthcare," *WIREs Data Mining and Knowledge Discovery*, vol. 12, no. 1, 2021.
[44] Y. Sun, H. Esaki, and H. Ochiai, "Visual analytics for anomaly classification in LAN based on deep convolutional neural network," in *2020 ICIEV and 2020 icIVPR*. IEEE, 2020.
[45] J. Wang, W. Zhang, H. Yang, C.-C. M. Yeh, and L. Wang, "Visual analytics for RNN-based deep reinforcement learning," *IEEE TVCG*, vol. 28, no. 12, pp. 4141–4155, 2022.
[46] T. Schreck, J. Bernard, T. Tekusova, and J. Kohlhammer, "Visual cluster analysis of trajectory data with interactive kohonen maps," in *2008 IEEE Symposium on Visual Analytics Science and Technology*. IEEE, oct 2008.
[47] B. C. Kwon, B. Eysenbach, J. Verma, K. Ng, C. D. Filippi, W. F. Stewart, and A. Perer, "Clustervision: Visual supervision of unsupervised clustering," *IEEE TVCG*, vol. 24, no. 1, pp. 142–151, jan 2018.
[48] D. Borland, W. Wang, J. Zhang, J. Shrestha, and D. Gotz, "Selection bias tracking and detailed subset comparison for high-dimensional data," 2019.
[49] T. Kohonen, *Self-Organizing Maps*. Springer, 2000.
[50] D. Sacha, M. Kraus, J. Bernard, M. Behrisch, T. Schreck, Y. Asano, and D. A. Keim, "SOMFlow: Guided exploratory cluster analysis with self-organizing maps and analytic provenance," *IEEE TVCG*, vol. 24, no. 1, pp. 120–130, 2018.
[51] M. Blazevic, L. B. Sina, D. Burkhardt, M. Siegel, and K. Nazemi, "Visual analytics and similarity search - interest-based similarity search in scientific data," in *International Conference Information Visualisation (IV)*. IEEE, jul 2021.
[52] J. Krause, A. Dasgupta, J. Swartz, Y. Aphinyanaphongs, and E. Bertini, "A workflow for visual diagnostics of binary classifiers using instance-level explanations," 2017.
[53] K. Nazemi, R. Retz, D. Burkhardt, A. Kuijper, J. Kohlhammer, and D. W. Fellner, "Visual trend analysis with digital libraries," in *Proceedings of the 15th i-KNOW Conference '15*. ACM Press, 2015.
[54] D. A. Keim, "Designing pixel-oriented visualization techniques: Theory and applications," *IEEE TVCG*, vol. 6, 2000.
[55] D. M. Blei, A. Y. Ng, and M. I. Jordan, "Latent dirichlet allocation," *Journal of Machine Learning Research*, vol. 3, 2003.
[56] W. He, L. Zou, A. K. Shekar, L. Gou, and L. Ren, "Where can we help? a visual analytics approach to diagnosing and improving semantic segmentation of movable objects," *IEEE TVCG*, vol. 28, no. 1, pp. 1040–1050, jan 2022.